

## 172. Physical Coding Sublayer (PCS), type 800GBASE-R

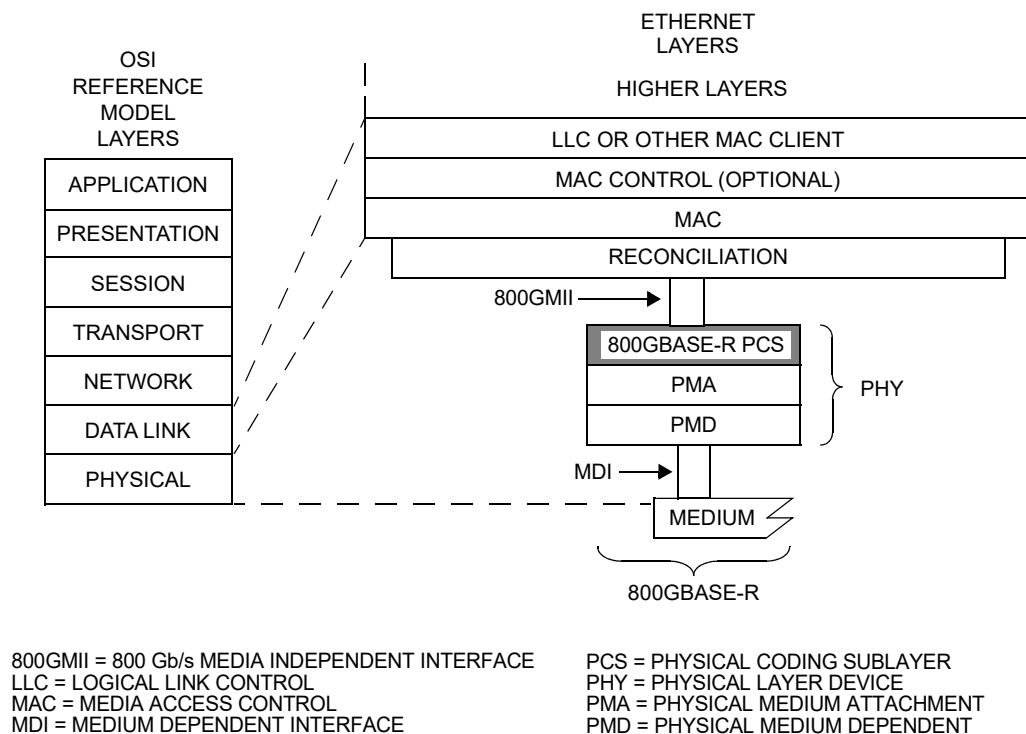
### 172.1 Overview

#### 172.1.1 Scope

This clause specifies the Physical Coding Sublayer (PCS) that is common to a family of Physical Layer implementations known as 800GBASE-R. The 800GBASE-R PCS is a sublayer of the 800 Gb/s PHYs listed in Table 169–1. The term 800GBASE-R is used when referring generally to Physical Layers using the PCS defined in this clause. Annex 172A provides examples of RS-FEC codewords constructed with the method specified in this clause.

#### 172.1.2 Relationship of the 800GBASE-R PCS to other standards

Figure 172–1 shows the relationship of the 800GBASE-R PCS with other sublayers to the ISO Open System Interconnection (OSI) reference model.



**Figure 172–1—800GBASE-R PCS relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model**

#### 172.1.3 Summary of functions

The PCS service interface is the 800 Gb/s Media Independent Interface (800GMII), which is defined in Clause 170. The 800GMII provides a uniform interface to the Reconciliation Sublayer for all 800 Gb/s PHY implementations.

The 800GBASE-R PCS is based on the 400GBASE-R PCS defined in [Clause 119](#).

The 800GBASE-R PCS provide all services required by the 800GMII, including the following:

- a) Encoding (decoding) of eight 800GMII data octets to (from) 66-bit blocks (64B/66B)
- b) Transcoding from (to) 66-bit blocks to (from) 257-bit blocks (256B/257B)
- c) Scrambling (descrambling) the 257-bit transcoded blocks
- d) Alignment marker insertion and removal
- e) Reed-Solomon encoding (decoding) the scrambled 257-bit blocks
- f) Distribution of Reed-Solomon encoded data to (from) PCS lanes
- g) Alignment marker lock, lane deskew and reordering of received PCS lanes
- h) Transferring encoded data to (from) the PMA
- i) Compensation for any rate differences caused by the insertion or deletion of alignment markers or due to any rate difference between the 800GMII and PMA through the insertion or deletion of idle control characters

#### 172.1.4 FEC Degrad

The FEC degrade feature provides the ability to detect and indicate a degrade condition at the RS-FEC decoder using FEC degrade detection, and to propagate the FEC degrade indication using FEC degrade signaling. The propagation of FEC degrade indications across PCS and XS is described in [116.6](#).

FEC degrade detection is specified in 172.2.5.3. FEC degrade detection is optional.

FEC degrade signaling using the PCS alignment marker status field is specified for the PCS in 172.2.4.6.

FEC degrade signaling across the PCS service interface is specified in 172.1.5.1.

#### 172.1.5 Inter-sublayer interfaces

The upper interface of the PCS connects to the Reconciliation Sublayer through the 800GMII. The lower interface of the PCS connects to the PMA to support a PMD. The 800GBASE-R PCS has a nominal rate at the PMA service interface of 26.5625 Gb/s on each of 32 PCS lanes, which provides capacity for the MAC data rate of 800 Gb/s.

While this specification defines interfaces in terms of bits, octets, and frames, implementations may choose other data-path widths for implementation convenience.

##### 172.1.5.1 PCS service interface

The PCS service interface allows the 800GBASE-R PCS to transfer information to and from a PCS client. The PCS client is either the Reconciliation Sublayer or the PHY 800GXS.

When the client sublayer is the Reconciliation Sublayer, the PCS service interface is the 800GMII defined in [Clause 170](#).

When the client sublayer is the PHY 800GXS, the PCS service interface is the 800GMII with additional signals TXRD, TXLD, RXRD, RXLD, and PCS\_status.

The TXRD and TXLD signals indicate the state of the remote degrade and local degrade, respectively, as detected by the PHY 800GXS in the transmit direction (see [171.3.3](#)).

The RXRD signal indicates the state of the rx\_rm\_degraded variable (see 172.2.6.2.2) as detected by the PCS in the receive direction.

The RXLD signal is the logical OR of the FEC\_degraded\_SER and rx\_local\_degraded variables (see 172.2.6.2.2) as detected by the PCS in the receive direction.

The PCS\_status signal indicates the state of the PCS\_status variable (see 119.2.6.2.2) as detected by the PCS in the receive direction.

#### **172.1.5.2 Physical Medium Attachment (PMA) service interface**

The PMA service interface supports the exchange of encoded data between the PCS and PMA. The PMA service interface is defined in 173.3 and is an instance of the inter-sublayer service interface definition in 169.3.

The PMA service interface for the PCS is described in an abstract manner and does not imply any particular implementation.

#### **172.1.6 Functional block diagram**

Figure 172–2 provides a functional block diagram of the 800GBASE-R PCS.

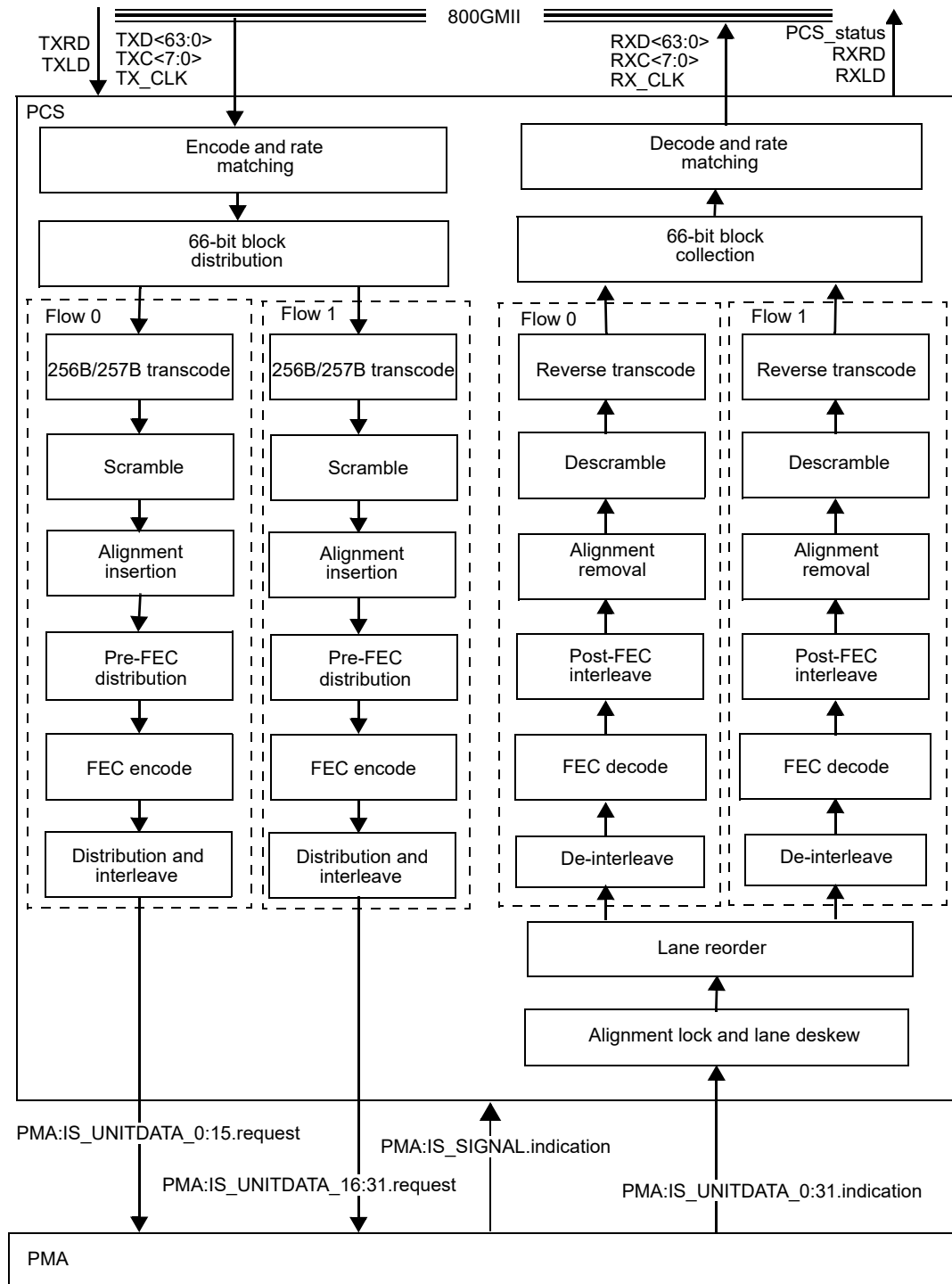


Figure 172-2—Functional block diagram

## 172.2 PCS functions

### 172.2.1 Overview

The 800GBASE-R PCS is composed of the PCS Transmit and PCS Receive processes. The PCS shields the Reconciliation Sublayer (and MAC) from the specific nature of the underlying channel.

When communicating with the 800GMII, the PCS uses an eight octet-wide, synchronous data path, with packet delineation being provided by transmit control signals ( $\text{TXC}_{<n>} = 1$ ) and receive control signals ( $\text{RXC}_{<n>} = 1$ ). When communicating with the PMA, the 800GBASE-R PCS uses 32 encoded bit streams also known as PCS lanes. Per direction (receive or transmit), the PCS lanes originate from a common clock but may vary in phase and skew dynamically. The PMA operates independently of block and packet boundaries. The PCS provides the functions necessary to map packets between the 800GMII format and the PMA service interface format.

The PCS transmit function can operate in normal mode or test-pattern mode.

When the transmit function is in normal mode, the PCS Transmit process continuously generates 66-bit blocks based on the  $\text{TXD}_{<63:0>}$  and  $\text{TXC}_{<7:0>}$  signals on the 800GMII. This takes place in the encode and rate matching function shown in Figure 172–2. The 66-bit blocks are then distributed between two parallel transmit functions, referred to as flow 0 and flow 1. Within each flow, the 66-bit blocks are transcoded to 257-bit blocks and scrambled. Alignment markers are periodically inserted into the data stream of each flow following the requirements specified in 172.2.4.6. For each flow, the data stream is distributed to two 5140-bit blocks and then FEC encoded. For each flow, the two FEC codewords are then interleaved before data is distributed to individual PCS lanes.

Transmit data units are sent to the service interface via the  $\text{PMA:IS\_UNITDATA\_0:31.request}$  primitive.

When the transmit function is in test-pattern mode, a test pattern is packed into the transmit data units that are sent to the PMA service interface via the  $\text{PMA:IS\_UNITDATA\_0:31.request}$  primitive.

In the receive direction, the PCS Synchronization process continuously monitors  $\text{PMA:IS\_SIGNAL.indication(SIGNAL\_OK)}$ . When  $\text{SIGNAL\_OK}$  indicates OK, then the PCS synchronization process accepts data units via the  $\text{PMA:IS\_UNITDATA\_0:31.indication}$  primitive. It attains alignment marker lock based on the common marker (CM) portion of the alignment markers that are periodically transmitted on every PCS lane and identifies individual PCS lanes using the unique marker (UM) portion of the alignment marker. The PCS lanes are then reordered and deskewed, and the  $\text{align\_status}$  flag is set. Note that a particular transmit PCS lane can be received on any receive lane of the service interface due to the skew and multiplexing that occurs in the path.

The PCS Receive process separates the reordered PCS lanes into two sets of 16 PCS lanes belonging to each flow. Within a flow, the data from the 16 PCS lanes is de-interleaved, processed by the FEC decoder, and re-interleaved on a 10-bit basis to form a single data stream. The alignment markers are removed, the data is descrambled and reverse transcoded back to 66-bit blocks. A 66-bit block collection function merges the 66-bit blocks from the two flows into a single stream of blocks that are then 64B/66B decoded.

### 172.2.2 Use of 66-bit blocks

The PCS maps 800GMII signals into 66-bit blocks, and vice versa, using a 64B/66B coding scheme. The PCS functions ENCODE and DECODE generate, manipulate, and interpret 66-bit blocks as defined in 172.2.3.

### 172.2.3 64B/66B code

The 64B/66B code is identical to that specified in 119.2.3 with the following exceptions:

- 200GMII/400GMII is replaced with 800GMII.
- 200GBASE-R/400GBASE-R is replaced with 800GBASE-R.
- EEE and Low Power Idle are not supported.

### 172.2.4 Transmit function

#### 172.2.4.1 64B/66B encoder

The transmit PCS generates 66-bit blocks based on the TXD<63:0> and TXC<7:0> signals received from the 800GMII. Each 800GMII transfer is encoded into one 66-bit block. The contents of each 66-bit block are contained in a vector tx\_coded<65:0> with tx\_coded<1:0> containing the sync header and the remainder of the bits the payload. The transmit PCS shall use the encoding method defined in either 172.2.4.1.1 or 172.2.4.1.2.

##### 172.2.4.1.1 State-diagram encoder

The state-diagram encoder generates 66-bit blocks based on the TXD<63:0> and TXC<7:0> signals received from the 800GMII as specified by the transmit state diagram shown in Figure 119–14.

##### 172.2.4.1.2 Stateless encoder

The stateless encoder generates 66-bit blocks based only on the current and preceding 800GMII transfers. Each 800GMII transfer is mapped into a 72-bit vector tx\_raw<71:0> (see 172.2.6.2.2). The encoder shall encode each tx\_raw<71:0> to a 66-bit block tx\_coded<65:0> according to the rules in Table 172–1. Constants LBLOCK\_T and EBLOCK\_T are defined in 172.2.6.2.1. Variables reset, tx\_raw, and tx\_coded are defined in 172.2.6.2.2. Functions T\_TYPE and ENCODE, and the block types, are defined in 172.2.6.2.3.

**Table 172–1—PCS stateless encoder rules**

| reset | T_TYPE(tx_raw <sub>i-1</sub> ) <sup>a</sup> | T_TYPE(tx_raw <sub>i</sub> ) <sup>b</sup> | Resulting tx_coded           |
|-------|---------------------------------------------|-------------------------------------------|------------------------------|
| 1     | any block type                              | any block type                            | LBLOCK_T                     |
| 0     | C or T                                      | C                                         | ENCODE(tx_raw <sub>i</sub> ) |
| 0     | C or T                                      | S                                         | ENCODE(tx_raw <sub>i</sub> ) |
| 0     | S or D                                      | D                                         | ENCODE(tx_raw <sub>i</sub> ) |
| 0     | S or D                                      | T                                         | ENCODE(tx_raw <sub>i</sub> ) |
| 0     | any combination not listed above            |                                           | EBLOCK_T                     |

<sup>a</sup> tx\_raw<sub>i-1</sub> is the 72-bit vector that immediately precedes tx\_raw<sub>i</sub>.

<sup>b</sup> tx\_raw<sub>i</sub> is the 72-bit vector that is being encoded.

#### 172.2.4.2 Rate matching

The transmit PCS may remove idle control characters or sequence ordered sets to compensate for the insertion of alignment markers. The transmit PCS may remove idle control characters or sequence ordered

sets or may insert idle control characters to compensate for different clock domains on the 800GMII and the PMA service interface. See 119.2.3.5 and 119.2.3.8 for the deletion and insertion rules.

#### 172.2.4.3 Block distribution

The 66-bit blocks are distributed to the two flows in an alternating fashion by the block distribution function such that the first 66-bit block is sent to flow 0, the second 66-bit block is sent to flow 1, the third 66-bit block is sent to flow 0, and subsequent 66-bit blocks continue the distribution procedure across the two flows.

#### 172.2.4.4 64B/66B to 256B/257B transcoder

The 64B/66B to 256B/257B transcoder in each flow is identical to that specified in 119.2.4.2.

NOTE—The two streams of 257-bit blocks generated by this process, together with the FEC\_degraded\_SER and rx\_local\_degraded bits should be used as the reference signal for mapping to OTN. The details of how to combine the two streams of 257-bit blocks into a single stream are outside the scope of this document.

#### 172.2.4.5 Scrambler

The scrambler in each flow is identical to that specified in 119.2.4.3. Although there is no requirement on the initial value of each scrambler, if an implementation sets the scrambler state to a fixed value (e.g., when reset is asserted), the two scramblers should be set to different states.

NOTE— If the two scramblers have the same state and the same input (e.g., encoded Remote Fault signal), their outputs will be identical. In that case, specific choices of input lanes in the PMA bit-multiplexing function can create atypical sequences on the PMA output.

#### 172.2.4.6 Alignment marker mapping and insertion

The alignment marker mapping and insertion in each flow is identical to the 400GBASE-R alignment marker and insertion function specified in 119.2.4.4 with the following exceptions:

- Alignment marker encoding values for flow 0 are specified in Table 172–2.
- Alignment marker encoding values for flow 1 are specified in Table 172–3 and the variable  $x$  in 119.2.4.4.2 takes the values of PCS lane number minus 16.
- The alignment marker insertion within each flow shall occur at the same point relative to the original stream of 66-bit blocks before block distribution. This requirement is illustrated in Figure 172–3 where the alignment marker group is inserted prior to the 257-bit block containing the 66-bit block  $k$  for flow 0 and prior to the 257-bit block containing the 66-bit block  $k+1$  for flow 1.
- When the client sublayer is the Reconciliation Sublayer, tx\_am\_sf<2:0> is assigned as follows:  
 $\text{tx\_am\_sf}\langle 2:0 \rangle = \{\text{FEC\_degraded\_SER} + \text{rx\_local\_degraded}, 0, 0\}$   
where FEC\_degraded\_SER and rx\_local\_degraded are defined in 172.2.6.2.2 and + denotes logical OR.
- When the client sublayer is the PHY 800GXS, tx\_am\_sf<2:0> is assigned as follows:  
 $\text{tx\_am\_sf}\langle 2:0 \rangle = \{\text{TXRD}, \text{TXLD}, 0\}$   
where TXRD and TXLD are signals from the PHY 800GXS on the PCS service interface (see 172.1.5.1).

The alignment marker format for each PCS lane is shown in Figure 119–4. There is a portion that is common across all alignment markers (designated as CM0 to CM5), a unique portion per PCS lane (designated as UM0 to UM5), and a unique pad per PCS lane within a flow (designated as UP0 to UP2). The pad in PCS lane  $j$  is the same as that in PCS lane  $j + 16$  where  $j$  takes values from 0 to 15.

NOTE—A text file containing the alignment marker patterns, as shown in Table 172–2 and Table 172–3, is available at <https://standards.ieee.org/downloads/802.3/>.

**Table 172–2—800GBASE-R flow 0 alignment marker encodings**

| PCS lane number | Encoding <sup>a</sup> |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|-----------------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|                 | CM0                   | CM1  | CM2  | UP0  | CM3  | CM4  | CM5  | UP1  | UM0  | UM1  | UM2  | UP2  | UM3  | UM4  | UM5  |
| 0               | 0x9A                  | 0x4A | 0x26 | 0xB6 | 0x65 | 0xB5 | 0xD9 | 0xD9 | 0xFE | 0x71 | 0xF3 | 0x26 | 0x01 | 0x8E | 0x0C |
| 1               |                       |      |      | 0x04 |      |      |      | 0x67 | 0xA5 | 0xDE | 0x7E | 0x98 | 0x5A | 0x21 | 0x81 |
| 2               |                       |      |      | 0x46 |      |      |      | 0xFE | 0xC1 | 0xF3 | 0x56 | 0x01 | 0x3E | 0x0C | 0xA9 |
| 3               |                       |      |      | 0x5A |      |      |      | 0x84 | 0x79 | 0x80 | 0xD0 | 0x7B | 0x86 | 0x7F | 0x2F |
| 4               |                       |      |      | 0xE1 |      |      |      | 0x19 | 0xD5 | 0x51 | 0xF2 | 0xE6 | 0x2A | 0xAE | 0x0D |
| 5               |                       |      |      | 0xF2 |      |      |      | 0x4E | 0xED | 0x4F | 0xD1 | 0xB1 | 0x12 | 0xB0 | 0x2E |
| 6               |                       |      |      | 0x3D |      |      |      | 0xEE | 0xBD | 0x9C | 0xA1 | 0x11 | 0x42 | 0x63 | 0x5E |
| 7               |                       |      |      | 0x22 |      |      |      | 0x32 | 0x29 | 0x76 | 0x5B | 0xCD | 0xD6 | 0x89 | 0xA4 |
| 8               |                       |      |      | 0x60 |      |      |      | 0x9F | 0x1E | 0x73 | 0x75 | 0x60 | 0xE1 | 0x8C | 0x8A |
| 9               |                       |      |      | 0x6B |      |      |      | 0xA2 | 0x8E | 0xC4 | 0x3C | 0x5D | 0x71 | 0x3B | 0xC3 |
| 10              |                       |      |      | 0xFA |      |      |      | 0x04 | 0x6A | 0xEB | 0xD8 | 0xFB | 0x95 | 0x14 | 0x27 |
| 11              |                       |      |      | 0x6C |      |      |      | 0x71 | 0xDD | 0x66 | 0x38 | 0x8E | 0x22 | 0x99 | 0xC7 |
| 12              |                       |      |      | 0x18 |      |      |      | 0x5B | 0x5D | 0xF6 | 0x95 | 0xA4 | 0xA2 | 0x09 | 0x6A |
| 13              |                       |      |      | 0x14 |      |      |      | 0xCC | 0xCE | 0x97 | 0xC3 | 0x33 | 0x31 | 0x68 | 0x3C |
| 14              |                       |      |      | 0xD0 |      |      |      | 0xB1 | 0x35 | 0xFB | 0xA6 | 0x4E | 0xCA | 0x04 | 0x59 |
| 15              |                       |      |      | 0xB4 |      |      |      | 0x56 | 0x59 | 0xBA | 0x79 | 0xA9 | 0xA6 | 0x45 | 0x86 |

<sup>a</sup> Each octet is transmitted LSB to MSB.

**Table 172–3—800GBASE-R flow 1 alignment marker encodings**

| PCS lane number | Encoding <sup>a</sup> |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|-----------------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|                 | CM0                   | CM1  | CM2  | UP0  | CM3  | CM4  | CM5  | UP1  | UM0  | UM1  | UM2  | UP2  | UM3  | UM4  | UM5  |
| 16              | 0x9A                  | 0x4A | 0x26 | 0xB6 | 0x65 | 0xB5 | 0xD9 | 0xD9 | 0x01 | 0x8E | 0x0C | 0x26 | 0xFE | 0x71 | 0xF3 |
| 17              |                       |      |      | 0x04 |      |      |      | 0x67 | 0x5A | 0x21 | 0x81 | 0x98 | 0xA5 | 0xDE | 0x7E |
| 18              |                       |      |      | 0x46 |      |      |      | 0xFE | 0x3E | 0x0C | 0xA9 | 0x01 | 0xC1 | 0xF3 | 0x56 |
| 19              |                       |      |      | 0x5A |      |      |      | 0x84 | 0x86 | 0x7F | 0x2F | 0x7B | 0x79 | 0x80 | 0xD0 |
| 20              |                       |      |      | 0xE1 |      |      |      | 0x19 | 0x2A | 0xAE | 0x0D | 0xE6 | 0xD5 | 0x51 | 0xF2 |
| 21              |                       |      |      | 0xF2 |      |      |      | 0x4E | 0x12 | 0xB0 | 0x2E | 0xB1 | 0xED | 0x4F | 0xD1 |
| 22              |                       |      |      | 0x3D |      |      |      | 0xEE | 0x42 | 0x63 | 0x5E | 0x11 | 0xBD | 0x9C | 0xA1 |
| 23              |                       |      |      | 0x22 |      |      |      | 0x32 | 0xD6 | 0x89 | 0xA4 | 0xCD | 0x29 | 0x76 | 0x5B |
| 24              |                       |      |      | 0x60 |      |      |      | 0x9F | 0xE1 | 0x8C | 0x8A | 0x60 | 0x1E | 0x73 | 0x75 |
| 25              |                       |      |      | 0x6B |      |      |      | 0xA2 | 0x71 | 0x3B | 0xC3 | 0x5D | 0x8E | 0xC4 | 0x3C |
| 26              |                       |      |      | 0xFA |      |      |      | 0x04 | 0x95 | 0x14 | 0x27 | 0xFB | 0x6A | 0xEB | 0xD8 |
| 27              |                       |      |      | 0x6C |      |      |      | 0x71 | 0x22 | 0x99 | 0xC7 | 0x8E | 0xDD | 0x66 | 0x38 |
| 28              |                       |      |      | 0x18 |      |      |      | 0x5B | 0xA2 | 0x09 | 0x6A | 0xA4 | 0x5D | 0xF6 | 0x95 |
| 29              |                       |      |      | 0x14 |      |      |      | 0xCC | 0x31 | 0x68 | 0x3C | 0x33 | 0xCE | 0x97 | 0xC3 |
| 30              |                       |      |      | 0xD0 |      |      |      | 0xB1 | 0xCA | 0x04 | 0x59 | 0x4E | 0x35 | 0xFB | 0xA6 |
| 31              |                       |      |      | 0xB4 |      |      |      | 0x56 | 0xA6 | 0x45 | 0x86 | 0xA9 | 0x59 | 0xBA | 0x79 |

<sup>a</sup> Each octet is transmitted LSB to MSB.

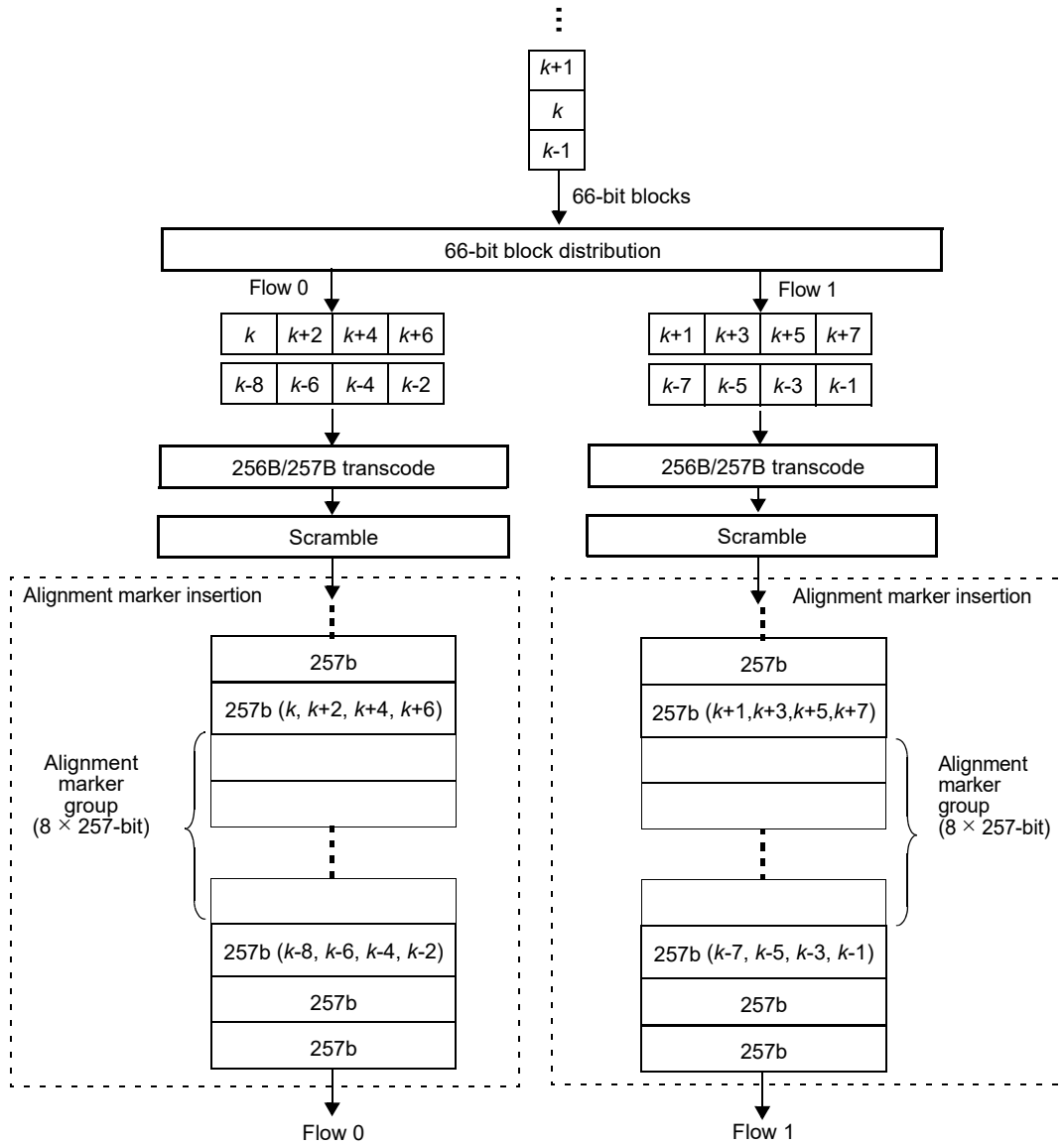


Figure 172-3—800GBASE-R PCS alignment marker insertion

#### 172.2.4.7 Pre-FEC distribution

The pre-FEC distribution in each flow is identical to that specified in 119.2.4.5.

#### 172.2.4.8 Reed-Solomon encoder

The Reed-Solomon encoder in each flow is identical to that specified in 119.2.4.6. Examples of RS-FEC codewords are provided in Annex 172A.

#### 172.2.4.9 Symbol distribution

The symbol distribution in each flow is identical to the symbol distribution of the 400GBASE-R PCS in 119.2.4.7.

#### 172.2.4.10 Transmit bit ordering and distribution

The transmit bit ordering and distribution is illustrated in Figure 172–4. The transmit bit ordering and distribution in “Flow 0 transmit function” and “Flow 1 transmit function” are identical to the 400GBASE-R transmit bit ordering and distribution in Figure 119–11 with the following exceptions:

- The “66-bit block distribution” of Figure 172–4 directly feeds into the “64B/66B to 256B/257B transcoder” in each flow’s transmit function shown in Figure 119–11, and the functional blocks above the “64B/66B to 256B/257B transcoder” in each flow are not used.
- For flow 1, the PMA service interface lane numbers 0:15 are replaced with 16:31, respectively.

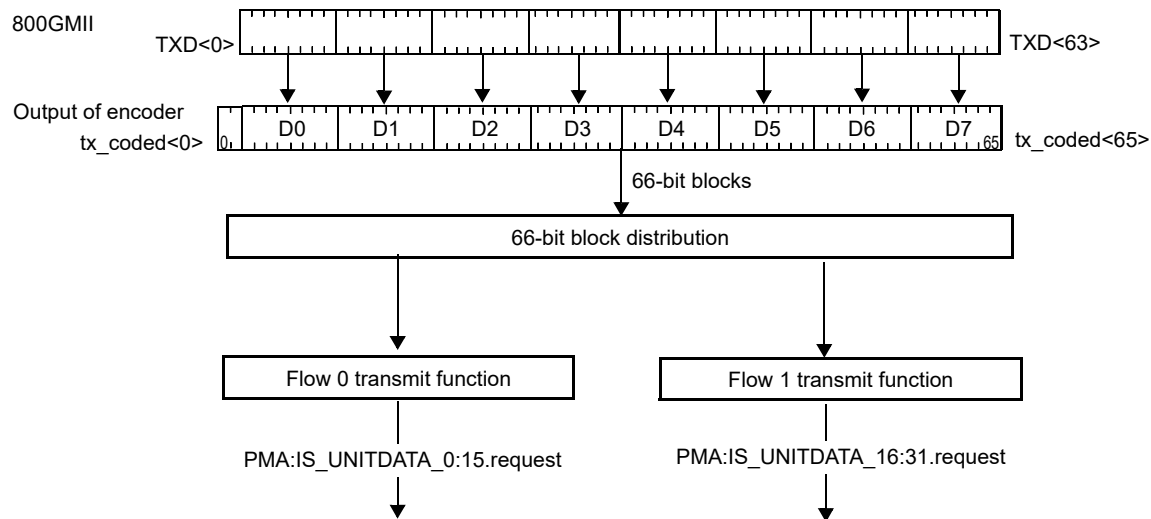


Figure 172–4—800GBASE-R PCS transmit bit ordering and distribution

#### 172.2.4.11 Test-pattern generator

The PCS shall have the ability to generate a scrambled idle test pattern, which is suitable for receiver tests and for certain transmitter tests. When test pattern generation is enabled (tx\_test\_mode is 1), the scrambled idle test pattern is generated by the PCS. The scrambled idle test pattern is the output of the PCS when the input to the PCS at the 800GMII is composed only of idle control characters.

If a Clause 45 MDIO is implemented, then the tx\_test\_mode variable is accessible through the register as shown in Table 172–5.

## 172.2.5 Receive function

### 172.2.5.1 Alignment lock and deskew

The alignment lock and deskew is identical to that specified in 119.2.5.1 with the following exceptions:

- $n = 32$
- Figure 119–13 is replaced with Figure 172–5 and Figure 172–6.
- The maximum Skew is changed from 180 ns to 152 ns.

### 172.2.5.2 Lane reorder and de-interleave

The PCS lanes might be received in any order on the PMA service interface lanes. The PCS receive function shall order the PCS lanes according to the PCS lane number. The PCS lane number is defined by the unique portion (UM<sub>0</sub> to UM<sub>5</sub>) of the alignment marker that is mapped to each PCS lane (see 172.2.4.6).

After all 32 PCS lanes are aligned, deskewed, and reordered, they are separated into two sets of 16 PCS lanes belonging to each flow. PCS lanes 0 to 15 are sent to flow 0 and PCS lanes 16 to 31 are sent to flow 1. Within a flow, the data from the 16 PCS lanes is de-interleaved to reconstruct the original two streams of FEC codewords.

### 172.2.5.3 Reed-Solomon decoder

The Reed-Solomon decoder is identical to that specified in 119.2.5.3, with the following exceptions:

- FEC\_degraded\_SER in flow 0 and flow 1 are mapped to FEC\_degraded\_SER\_0 and FEC\_degraded\_SER\_1, respectively.
- hi\_ser in flow 0 and flow 1 are mapped to hi\_ser\_0 and hi\_ser\_1, respectively.

The FEC degrade detection is optional.

### 172.2.5.4 Post-FEC interleave

The post-FEC interleave in each flow is identical to that specified in 119.2.5.4.

### 172.2.5.5 Alignment marker removal

The alignment marker removal in each flow is identical to that of the 400GBASE-R PCS in 119.2.5.5.

### 172.2.5.6 Descrambler

The descrambler in each flow is identical to that specified in 119.2.5.6.

### 172.2.5.7 256B/257B to 64B/66B transcoder

The 256B/257B to 64B/66B transcoder in each flow is identical to that specified in 119.2.5.7.

### 172.2.5.8 Block collection

The block collection reverses the block distribution done in the transmitter (see 172.2.4.3) by combining the 66-bit blocks from the two flows in an alternating fashion to form a single stream of 66-bit blocks. The first 66-bit block after the alignment marker group from flow 0 shall be followed by the first 66-bit block after the alignment marker group from flow 1.

### 172.2.5.9 64B/66B decoder

The receive PCS decodes 66-bit blocks to produce RXD<63:0> and RXC<7:0> for transmission to the 800GMII. One 800GMII transfer is decoded from each 66-bit block. The receive PCS shall use one of the two decoding methods that are defined in 172.2.5.9.1 and 172.2.5.9.2.

#### 172.2.5.9.1 State-diagram decoder

The state-diagram decoder decodes 66-bit blocks to produce RXD<63:0> and RXC<7:0> for transmission to the 800GMII as specified by the receive state diagram shown in Figure 119–15.

#### 172.2.5.9.2 Stateless decoder

The stateless decoder generates 800GMII transfers based only on the current and preceding 66-bit blocks. The decoder shall decode each 66-bit block rx\_coded<65:0> to a 72-bit vector rx\_raw<71:0> (see 172.2.6.2.2) according to the rules in Table 172–4. Constants LBLOCK\_R and EBLOCK\_R are defined in 172.2.6.2.1. Variables reset, rx\_raw, and rx\_coded are defined in 172.2.6.2.2. Functions R\_TYPE and DECODE, and the block types, are defined in 172.2.6.2.3.

**Table 172–4—PCS stateless decoder rules**

| reset | R_TYPE(rx_coded <sub>i-1</sub> ) <sup>a</sup> | R_TYPE(rx_coded <sub>i</sub> ) <sup>b</sup> | Resulting rx_raw               |
|-------|-----------------------------------------------|---------------------------------------------|--------------------------------|
| 1     | any block type                                | any block type                              | LBLOCK_R                       |
| 0     | any block type                                | E                                           | EBLOCK_R                       |
| 0     | E                                             | any block type                              | EBLOCK_R                       |
| 0     | any combination not listed above              |                                             | DECODE(rx_coded <sub>i</sub> ) |

<sup>a</sup> rx\_coded<sub>i-1</sub> is the 66-bit block that immediately precedes rx\_coded<sub>i</sub>.

<sup>b</sup> rx\_coded<sub>i</sub> is the 66-bit block that is being decoded.

### 172.2.5.10 Rate matching

The receive PCS may insert idle control characters to compensate for the deletion of alignment markers. The receive PCS may remove idle control characters or sequence ordered sets or may insert idle control characters to compensate for different clock domains on the PMA service interface and the 800GMII. See 119.2.3.5 and 119.2.3.8 for the deletion and insertion rules.

NOTE—If no idle control characters are inserted to compensate for the removal of alignment markers, the average data rate on the 800GMII will be reduced by approximately 49 ppm.

## 172.2.6 Detailed functions and state diagrams

### 172.2.6.1 State diagram conventions

The body of this subclause is composed of state diagrams, including the associated definitions of variables, functions, and counters. Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

The notation used in the state diagrams follows the conventions of 21.5. The notation ++ after a counter or integer variable indicates that its value is to be incremented by 1.

## 172.2.6.2 State variables

### 172.2.6.2.1 Constants

The constants are the same as those specified in 119.2.6.2.1 with the following exception:

- 200GMII/400GMII is replaced with 800GMII.

### 172.2.6.2.2 Variables

The variables are the same as those specified in 119.2.6.2.2 with the following exceptions:

- 200GMII/400GMII is replaced with 800GMII.
- Some variables are modified or added as follows.

The following variable definitions are used in place of those defined in 119.2.6.2.2.

amps\_lock<x>

Same behavior as in 119.2.6.2.2 with  $x = 0:31$ .

current\_pcsI

Same behavior as in 119.2.6.2.2 with the exception that the unique marker bits are defined in Table 172–2 and Table 172–3.

pcs\_alignment\_valid

Same behavior as 119.2.6.2.2 with the exception that the unique alignment markers are defined in 172.2.4.6.

restart\_lock

Boolean variable that is set by the PCS synchronization process to restart the alignment marker lock process on all PCS lanes. It is set to true in the DESKEW\_FAIL state or if either restart\_lock<y> are true. It is set to false upon entry into the LOSS\_OF\_ALIGNMENT state.

rx\_local\_degraded

Boolean variable that is set to true when the receiver in either flow detects rx\_am\_sf<1> asserted for two consecutive alignment marker periods. It is deasserted when rx\_am\_sf<1> is deasserted for two consecutive alignment marker periods in both flows. If a Clause 45 MDIO is implemented, this variable is mapped per Table 172–6.

rx\_rm\_degraded

Boolean variable that is set to true when the receiver in either flow detects rx\_am\_sf<2> asserted for two consecutive alignment marker periods. It is deasserted when rx\_am\_sf<2> is deasserted for two consecutive alignment marker periods in both flows. If a Clause 45 MDIO is implemented, this variable is mapped per Table 172–6.

The following variable definitions are added to those defined in 119.2.6.2.2.

FEC\_degraded\_SER

Boolean variable that is the logical OR of FEC\_degraded\_SER\_0 and FEC\_degraded\_SER\_1 (see 172.2.5.3). If a Clause 45 MDIO is implemented, the status of this variable is mapped per Table 172–6.

hi\_ser\_combined

Boolean variable that is the logical OR of hi\_ser\_0 and hi\_ser\_1 (see 172.2.5.3).  
If a Clause 45 MDIO is implemented, the status of this variable is mapped per  
Table 172–6.

restart\_lock<y>

Per flow Boolean variable that is set to true after 3 consecutive uncorrected  
codewords are received (3\_BAD state) by the RS-FEC decoder of flow *y*, where  
*y* is 0 or 1.

### 172.2.6.2.3 Functions

The functions are the same as those specified in 119.2.6.2.3 with the following exceptions:

- 200GMII/400GMII is replaced with 800GMII.
- EEE and Low Power Idle are not supported.

### 172.2.6.2.4 Counters

The counters are the same as those specified in 119.2.6.2.4 for the 400GBASE-R PCS with the following  
exception:

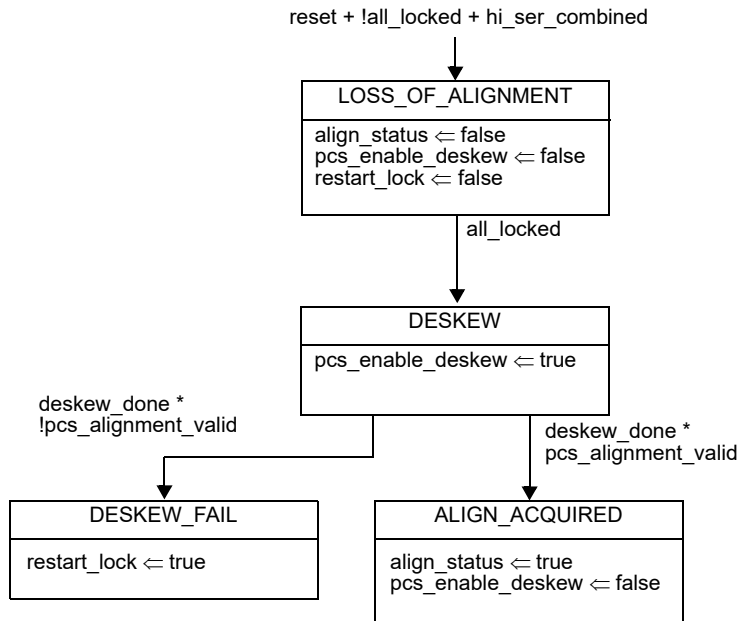
amp\_counter

This counter counts the interval of 16 384 FEC codewords containing alignment  
marker payload sequences for the 800GBASE-R PCS.

### 172.2.6.3 State diagrams

The state diagrams are identical to those specified in 119.2.6.3 with the following exceptions:

- The 800GBASE-R PCS implements 32 alignment marker lock processes (Figure 119–12).
- The PCS lane mapping registers are found in MDIO registers 3.400 to 3.431 (see 172.3.1).
- The PCS synchronization process is depicted in Figure 172–5 and Figure 172–6, instead of in  
Figure 119–13.
- The monitor for three consecutive uncorrectable FEC codewords (see Figure 172–6) is done  
independently within each flow.



**Figure 172–5—PCS synchronization state diagram**

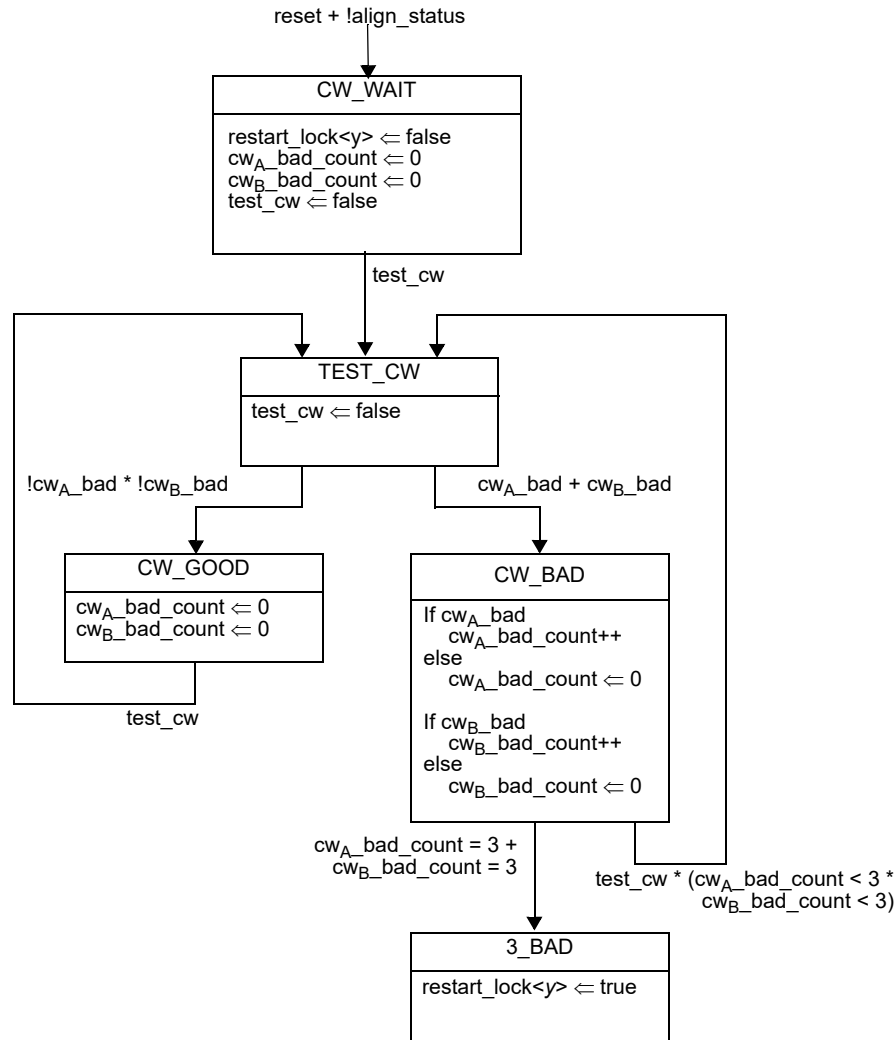


Figure 172-6—PCS per-flow codeword monitor state diagram

## 172.3 PCS management

The following objects apply to PCS management. If an MDIO Interface is provided (see Clause 45), they are accessed via that interface. If not, it is recommended that an equivalent access is provided.

### 172.3.1 PCS MDIO function mapping

The optional MDIO capability described in Clause 45 defines several registers that provide control and status information for and about the PCS. If MDIO is implemented, it shall map MDIO control bits to PCS control variables, as shown in Table 172-5, and MDIO status bits to PCS status variables, as shown in Table 172-6. MDIO registers relevant to the 800GBASE-R PCS are described in 45.2.3.

**Table 172–5—MDIO/PCS control variable mapping**

| MDIO control variable                     | PCS register name                                  | Register/bit number | PCS control variable                  |
|-------------------------------------------|----------------------------------------------------|---------------------|---------------------------------------|
| Loopback                                  | PCS control 1 register                             | 3.0.14              | Loopback                              |
| Reset                                     | PCS control 1 register                             | 3.0.15              | reset                                 |
| Transmit test-pattern enable              | BASE-R PCS test-pattern control register           | 3.42.3              | tx_test_mode                          |
| PCS FEC bypass indication enable          | PCS FEC control register                           | 3.800.1             | FEC_bypass_indication_enable          |
| PCS FEC degraded SER enable               | PCS FEC control register                           | 3.800.2             | FEC_degraded_SER_enable               |
| PCS FEC degraded SER activate threshold   | PCS FEC degraded SER activate threshold register   | 3.806, 3.807        | FEC_degraded_SER_activate_threshold   |
| PCS FEC degraded SER deactivate threshold | PCS FEC degraded SER deactivate threshold register | 3.808, 3.809        | FEC_degraded_SER_deactivate_threshold |
| PCS FEC degraded SER interval             | PCS FEC degraded SER interval                      | 3.810, 3.811        | FEC_degraded_SER_interval             |

**Table 172–6—MDIO/PCS status variable mapping**

| MDIO status variable                        | PCS register name                                    | Register/bit number               | PCS status variable             |
|---------------------------------------------|------------------------------------------------------|-----------------------------------|---------------------------------|
| BASE-R and MultiGBASE-T receive link status | BASE-R and MultiGBASE-T PCS status 1 register        | 3.32.12                           | PCS_status                      |
| PCS lane alignment status                   | Multi-lane BASE-R PCS alignment status 1 register    | 3.50.12                           | align_status                    |
| Lane 0 to 31 aligned                        | Multi-lane BASE-R PCS alignment status registers     | 3.52.7:0<br>3.53.15:0<br>3.54.7:0 | amps_lock<0:31>                 |
| FEC codeword counter                        | PCS FEC codeword counter register                    | 3.300 to 3.302                    | FEC_cw_counter                  |
| FEC codeword error bin counter, 1 to 15     | PCS FEC codeword error bin, 1 to 15                  | 3.340 to 3.369                    | FEC_codeword_error_bin_<1:15>   |
| Lane 0 to 31 mapping                        | Lane <i>x</i> mapping register                       | 3.400 through 3.431               | pcs_lane_mapping<0:31>          |
| PCS FEC symbol errors, PCS lanes 0 to 31    | PCS FEC symbol error counter register, lanes 0 to 31 | 3.600 to 3.663                    | FEC_symbol_error_counter_<0:31> |
| PCS FEC bypass indication ability           | PCS FEC status register                              | 3.801.1                           | FEC_bypass_indication_ability   |
| PCS FEC high SER                            | PCS FEC status register                              | 3.801.2                           | hi_ser_combined                 |
| PCS FEC degraded SER ability                | PCS FEC status register                              | 3.801.3                           | FEC_degraded_SER_ability        |
| PCS FEC degraded SER                        | PCS FEC status register                              | 3.801.4                           | FEC_degraded_SER                |

**Table 172–6—MDIO/PCS status variable mapping (*continued*)**

| MDIO status variable          | PCS register name                              | Register/bit number | PCS status variable        |
|-------------------------------|------------------------------------------------|---------------------|----------------------------|
| Remote degraded SER received  | PCS FEC status register                        | 3.801.5             | rx_rm_degraded             |
| Local degraded SER received   | PCS FEC status register                        | 3.801.6             | rx_local_degraded          |
| PCS FEC corrected codewords   | PCS FEC corrected codewords counter register   | 3.802, 3.803        | FEC_corrected_cw_counter   |
| PCS FEC uncorrected codewords | PCS FEC uncorrected codewords counter register | 3.804, 3.805        | FEC_uncorrected_cw_counter |

### 172.3.2 FEC\_corrected\_cw\_counter

The FEC\_corrected\_cw\_counter is identical to 119.3.2 with the clarification that the count includes both flows.

### 172.3.3 FEC\_uncorrected\_cw\_counter

The FEC\_uncorrected\_cw\_counter is identical to 119.3.3 with the clarification that the count includes both flows.

### 172.3.4 FEC\_symbol\_error\_counter<sub>*i*</sub>

FEC\_symbol\_error\_counter<sub>*i*</sub> (where *i*=0 to 31 for the 800GBASE-R PCS) are 32-bit counters that count once for each 10-bit symbol corrected on PCS lane *i* when align\_status is true. These variables are mapped to the registers defined in Table 172–6.

### 172.3.5 FEC\_cw\_counter

The FEC\_cw\_counter is an optional 48-bit counter that counts once for each FEC codeword received when align\_status is true. This variable is mapped to the registers defined in 45.2.3.48a (3.300 to 3.302).

### 172.3.6 FEC\_codeword\_error\_bin<sub>*i*</sub>

FEC\_codeword\_error\_bin<sub>*i*</sub>, where *i*=1 to 15, are optional 32-bit counters. While align\_status is true, for each codeword received with exactly *i* correctable 10-bit symbols, FEC\_codeword\_error\_bin<sub>*i*</sub> is incremented. For example, if a codeword has exactly five errored 10-bit symbols, then fec\_codeword\_error\_bin\_5 is incremented. These variables are mapped to registers defined in 45.2.3.48b (3.340 to 3.369).

## 172.4 Loopback

Loopback is identical to that specified in 119.4.

## 172.5 Delay constraints

The maximum delay contributed by the 800GBASE-R PCS (sum of transmit and receive delays at one end of the link) shall be no more than 640 000 BT (1250 pause\_quanta or 800 ns). A description of overall system delay constraints and the definitions for bit times and pause\_quanta can be found in 169.4 and its references.

## 172.6 Auto-Negotiation

The following requirements apply to a PCS used with an 800GBASE-CR8 or 800GBASE-KR8 PMD where support for the Auto-Negotiation process defined in Clause 73 is mandatory. The PCS shall support the AN\_LINK.indication(link\_status) primitive (see 73.9). The parameter link\_status shall take the value FAIL when PCS\_status=false and the value OK when PCS\_status=true. The primitive shall be generated when the value of link\_status changes.

## 172.7 Protocol implementation conformance statement (PICS) proforma for Clause 172, Physical Coding Sublayer (PCS), type 800GBASE-R<sup>13</sup>

### 172.7.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 172, Physical Coding Sublayer (PCS), type 800GBASE-R, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

### 172.7.2 Identification

#### 172.7.2.1 Implementation identification

|                                                                                                                                                                                                                                                                                          |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Supplier <sup>1</sup>                                                                                                                                                                                                                                                                    |  |
| Contact point for inquiries about the PICS <sup>1</sup>                                                                                                                                                                                                                                  |  |
| Implementation Name(s) and Version(s) <sup>1,3</sup>                                                                                                                                                                                                                                     |  |
| Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) <sup>2</sup>                                                                                                                                      |  |
| NOTE 1—Required for all implementations.<br>NOTE 2—May be completed as appropriate in meeting the requirements for the identification.<br>NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model). |  |

#### 172.7.2.2 Protocol summary

|                                                                                                                                                                    |                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Identification of protocol standard                                                                                                                                | IEEE Std 802.3df-2024, Clause 172, Physical Coding Sublayer (PCS), type 800GBASE-R |
| Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS                                                    |                                                                                    |
| Have any Exception items been required? No [ ] Yes [ ]<br>(See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-2024.) |                                                                                    |

|                   |  |
|-------------------|--|
| Date of Statement |  |
|-------------------|--|

<sup>13</sup>Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

### 172.7.3 Major capabilities/options

| Item | Feature                                | Subclause | Value/Comment | Status | Support           |
|------|----------------------------------------|-----------|---------------|--------|-------------------|
| MII  | 800GMII logical interface is supported | 172.1.5.1 |               | O      | Yes [ ]<br>No [ ] |
| *MD  | MDIO                                   | 172.3.1   |               | O      | Yes [ ]<br>No [ ] |
| *FDD | Support for FEC degrade detection      | 172.2.5.3 |               | O      | Yes [ ]<br>No [ ] |

### 172.7.4 PICS proforma tables for Physical Coding Sublayer (PCS), type 800GBASE-R

#### 172.7.4.1 Transmit function

| Item | Feature                                                                      | Subclause   | Value/Comment | Status | Support            |
|------|------------------------------------------------------------------------------|-------------|---------------|--------|--------------------|
| *SE  | Uses stateless encoder                                                       | 172.2.4.1.2 |               | O      | Yes [ ]<br>No [ ]  |
| TF1  | Transmit 64B/66B encoder uses state diagram in <a href="#">Figure 119–14</a> | 172.2.4.1.1 |               | !SE:M  | Yes [ ]<br>N/A [ ] |
| TF2  | Transmit 64B/66B encoder uses stateless rules in Table 172–1                 | 172.2.4.1.2 |               | SE:M   | Yes [ ]<br>N/A [ ] |
| TF3  | 64B/66B to 256B/257B transcoder                                              | 172.2.4.4   |               | M      | Yes [ ]            |
| TF4  | Scrambler                                                                    | 172.2.4.5   |               | M      | Yes [ ]            |
| TF5  | Pad value                                                                    | 172.2.4.6   |               | M      | Yes [ ]            |
| TF6  | Alignment marker insertion                                                   | 172.2.4.6   |               | M      | Yes [ ]            |
| TF7  | FEC degrade signaling                                                        | 172.2.4.6   |               | M      | Yes [ ]            |
| TF8  | Pre-FEC distribution                                                         | 172.2.4.7   |               | M      | Yes [ ]            |
| TF9  | Reed-Solomon encoder                                                         | 172.2.4.8   |               | M      | Yes [ ]            |
| TF10 | Symbol distribution                                                          | 172.2.4.9   |               | M      | Yes [ ]            |
| TF11 | Transmission bit ordering                                                    | 172.2.4.10  |               | M      | Yes [ ]            |
| TF12 | Scrambled idle transmit test-pattern generator                               | 172.2.4.11  |               | M      | Yes [ ]            |

### 172.7.4.2 Receive function

| Item | Feature                                                                                   | Subclause   | Value/Comment | Status | Support            |
|------|-------------------------------------------------------------------------------------------|-------------|---------------|--------|--------------------|
| *BI  | Bypass indication supported                                                               | 172.2.5.3   |               | O      | Yes [ ]<br>No [ ]  |
| *SD  | Uses stateless decoder                                                                    | 172.2.5.9.2 |               | O      | Yes [ ]<br>No [ ]  |
| RF1  | Alignment marker lock                                                                     | 172.2.5.1   |               | M      | Yes [ ]            |
| RF2  | PCS lane number is captured to an MDIO register                                           | 172.2.6.3   |               | MD:M   | Yes [ ]<br>N/A [ ] |
| RF3  | Deskew                                                                                    | 172.2.5.1   |               | M      | Yes [ ]            |
| RF4  | Lane reorder and de-interleave                                                            | 172.2.5.2   |               | M      | Yes [ ]            |
| RF5  | Reed-Solomon decoder corrects any combination of up to $t=15$ symbol errors in a codeword | 172.2.5.3   |               | M      | Yes [ ]            |
| RF6  | Reed-Solomon decoder is capable of indicating when a codeword was not corrected           | 172.2.5.3   |               | M      | Yes [ ]            |
| RF7  | Error monitoring and assertion of hi_ser, while error indication is bypassed              | 172.2.5.3   |               | BI:M   | Yes [ ]<br>N/A [ ] |
| RF8  | Detect FEC degrade                                                                        | 172.2.5.3   |               | FDD:M  | Yes [ ]<br>N/A [ ] |
| RF9  | Alignment marker removal                                                                  | 172.2.5.5   |               | M      | Yes [ ]            |
| RF10 | Descrambler                                                                               | 172.2.5.6   |               | M      | Yes [ ]            |
| RF11 | 256B/257B to 64B/66B transcoder                                                           | 172.2.5.7   |               | M      | Yes [ ]            |
| RF12 | Receive 64B/66B decoder uses state diagram in <a href="#">Figure 119–15</a>               | 172.2.5.9.1 |               | !SD:M  | Yes [ ]<br>N/A [ ] |
| RF13 | Receive 64B/66B decoder uses stateless rules in Table 172–4                               | 172.2.5.9.2 |               | SD:M   | Yes [ ]<br>N/A [ ] |

### 172.7.4.3 64B/66B coding rules

| Item | Feature                                                        | Subclause            | Value/Comment                                                                      | Status | Support |
|------|----------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------|--------|---------|
| C1   | Encoder (and ENCODE function) implements the code as specified | 172.2.3, 172.2.6.2.3 |                                                                                    | M      | Yes [ ] |
| C2   | Decoder (and DECODE function) implements the code as specified | 172.2.3, 172.2.6.2.3 |                                                                                    | M      | Yes [ ] |
| C3   | Only valid block types are transmitted                         | 172.2.3              |                                                                                    | M      | Yes [ ] |
| C4   | Invalid block types are treated as an error                    | 172.2.3              |                                                                                    | M      | Yes [ ] |
| C5   | Only valid control characters are transmitted                  | 172.2.3              |                                                                                    | M      | Yes [ ] |
| C6   | Invalid control characters are treated as an error             | 172.2.3              |                                                                                    | M      | Yes [ ] |
| C7   | Idles do not interrupt data                                    | 172.2.3              |                                                                                    | M      | Yes [ ] |
| C8   | IDLE control code insertion and deletion                       | 172.2.3              | Insertion or deletion in groups of 8 /I/s                                          | M      | Yes [ ] |
| C9   | Sequence ordered set deletion                                  | 172.2.3              | Only one whole ordered set of two consecutive sequence ordered sets may be deleted | M      | Yes [ ] |

### 172.7.4.4 Management

| Item | Feature                                                 | Subclause | Value/Comment | Status | Support                      |
|------|---------------------------------------------------------|-----------|---------------|--------|------------------------------|
| M1   | Equivalent access to PCS management objects is provided | 172.3     |               | !MD:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| M2   | Mapping of MDIO control bits and MDIO status bits       | 172.3.1   |               | MD:M   | Yes [ ]<br>N/A [ ]           |

### 172.7.4.5 Loopback

| Item | Feature                                                       | Subclause | Value/Comment | Status | Support |
|------|---------------------------------------------------------------|-----------|---------------|--------|---------|
| L1   | Supports loopback                                             | 172.4     |               | M      | Yes [ ] |
| L2   | When in loopback, transmits what it receives from the 800GMII | 172.4     |               | M      | Yes [ ] |
| L3   | When in loopback, ignore all data presented by the PMA        | 172.4     |               | M      | Yes [ ] |

#### 172.7.4.6 Delay constraints

| Item | Feature              | Subclause | Value/Comment | Status | Support |
|------|----------------------|-----------|---------------|--------|---------|
| TIM1 | PCS delay constraint | 172.5     |               | M      | Yes [ ] |

#### 172.7.4.7 Auto-Negotiation

| Item | Feature                                                  | Subclause | Value/Comment                                            | Status | Support            |
|------|----------------------------------------------------------|-----------|----------------------------------------------------------|--------|--------------------|
| *AN  | Support for use with an 800GBASE-CR8 or 800GBASE-KR8 PMD | 172.6     | AN technology dependent interface described in Clause 73 | O      | Yes [ ]<br>No [ ]  |
| AN1  | AN_LINK.indication primitive                             | 172.6     | Support of the AN_LINK.indication(link_status) primitive | AN:M   | Yes [ ]<br>N/A [ ] |
| AN2  | link_status parameter                                    | 172.6     | Takes the value OK or FAIL                               | AN:M   | Yes [ ]<br>N/A [ ] |
| AN3  | Generation of AN_LINK.indication primitive               | 172.6     | Generated when the value of link_status changes          | AN:M   | Yes [ ]<br>N/A [ ] |